Chapter 11 Mid-Chapter Quiz

Quadratic Functions

FangMath

Name: _____ Date: _____

Level 1 — Basic Concepts

Part A: Identify the Vertex and Axis of Symmetry

Graph each function. Label the **axis of symmetry** and the **vertex**.

1. $y = \frac{1}{2}x^2$

2. $y = -2x^2 - 1$

3. $y = 3x^2 - 6x$

4. $y = x^2 + 2x + 4$

5. $y = -0.5x^2 + 2x + 1$

Part B: Order by Width of Graph

Order each group of quadratic functions from **widest** to **narrowest** graph.

6. $y = 2x^2, \quad y = 0.5x^2, \quad y = -x^2$

Answer: _____

7. $f(x) = 4x^2, \quad f(x) = \frac{2}{3}x^2, \quad f(x) = 3x^2$

Answer: _____

8. $f(x) = 0.6x^2, \quad f(x) = 0.3x^2, \quad f(x) = 0.2x^2$

Answer: _____

9. $y = -2x^2, \quad y = x^2, \quad y = -0.25x^2$

Answer: _____

Level 2 — Solving Quadratics

Part C: Solve by Finding Square Roots

Solve each equation by finding square roots. If the equation has **no real-number solution**, write *no solution*.

10. $m^2 = 81$

11. $5r^2 - 180 = 0$

12. $t^2 - 7 = -18$

13. $36n^2 = 9$

Part D: Solve by Graphing

Solve each equation by **graphing the related function**. If the equation has no real-number solution, write *no solution*.

14. $x^2 - 16 = 0$

15. $x^2 + 9 = 0$

16. $0.25x^2 = 0$

Part E: Solve by Factoring

Solve each equation by **factoring**.

17. $b^2 + 3b - 4 = 0$

18. $n^2 + n - 12 = 0$

19. $2x^2 - 5x - 3 = 0$

20. $t^2 - 3t = 28$

21. $3n^2 = 6n$

Level 3 — Advanced & Word Problems

Part F: Conceptual Questions

22. Writing. Describe the steps you would use to graph the function $y = 2x^2 + 5$.

23. Reasoning. Does the value of c in the quadratic function $y = ax^2 + bx + c$ affect the *horizontal position* of the vertex of the graph? Explain why or why not.

24. Writing. Describe how the graph of $y = 3x^2$ differs from the graph of $y = x^2$.

Part G: Open-Ended

Give an example of a quadratic function that matches each description.

25. The axis of symmetry is to the **left** of the y-axis.

$f(x) =$ _____

26. Its graph lies **entirely below** the x-axis.

$f(x) =$ _____

27. Its graph opens **upward** and has its vertex at $(0, 0)$.

$f(x) =$ _____

Part H: Extended Problem

28. **a.** Solve $x^2 - 4 = 0$ and $2x^2 - 8 = 0$ by graphing their related functions.

b. Reasoning. Why does it make sense that the graphs have the same x-intercepts?

29. **(STEM — Construction)** You are building a rectangular planter for your school garden. You want the area of the bottom to be 90 ft^2. You want the length of the planter to be 3 ft longer than *twice* its width. What should the dimensions of the bottom of the planter be?

30. (Sewing) You have 324 ft^2 of fabric to make a circular play parachute for kids. What is the radius of the largest parachute you could make? Round to the nearest tenth of a foot. *(Hint: $A = \pi r^2$)*

Answer Key

Answer Key

Part A — Graphs (vertex and axis of symmetry)

1. $y = \frac{1}{2}x^2$: vertex $(0, 0)$, axis $x = 0$. Opens upward, wider than $y = x^2$.

2. $y = -2x^2 - 1$: vertex $(0, -1)$, axis $x = 0$. Opens downward.

3. $y = 3x^2 - 6x = 3(x - 1)^2 - 3$: vertex $(1, -3)$, axis $x = 1$.

4. $y = x^2 + 2x + 4 = (x + 1)^2 + 3$: vertex $(-1, 3)$, axis $x = -1$.

5. $y = -0.5x^2 + 2x + 1 = -0.5(x - 2)^2 + 3$: vertex $(2, 3)$, axis $x = 2$.

Part B — Widest to Narrowest (smaller $|a|$ = wider graph)

6. $y = 0.5x^2,\ y = -x^2,\ y = 2x^2$

7. $f(x) = \frac{2}{3}x^2,\ f(x) = 3x^2,\ f(x) = 4x^2$

8. $f(x) = 0.2x^2,\ f(x) = 0.3x^2,\ f(x) = 0.6x^2$

9. $y = -0.25x^2,\ y = x^2,\ y = -2x^2$

Part C — Square Roots

10. $m^2 = 81 \Rightarrow m = \pm 9$

11. $5r^2 = 180 \Rightarrow r^2 = 36 \Rightarrow r = \pm 6$

12. $t^2 = -11$: *no solution* (negative under square root)

13. $36n^2 = 9 \Rightarrow n^2 = \frac{1}{4} \Rightarrow n = \pm\frac{1}{2}$

Part D — Graphing

14. $x^2 = 16 \Rightarrow x = \pm 4$

15. $x^2 = -9$: *no solution*

16. $0.25x^2 = 0 \Rightarrow x = 0$

Part E — Factoring

17. $(b + 4)(b - 1) = 0 \Rightarrow b = -4,\ 1$

18. $(n + 4)(n - 3) = 0 \Rightarrow n = -4,\ 3$

19. $(2x+1)(x-3)=0 \Rightarrow x=-\frac{1}{2},\ 3$

20. $t^2-3t-28=0 \Rightarrow (t-7)(t+4)=0 \Rightarrow t=7,\ -4$

21. $3n^2-6n=0 \Rightarrow 3n(n-2)=0 \Rightarrow n=0,\ 2$

Part F — Conceptual

22. Graph $y=2x^2+5$: vertex at $(0,5)$; axis of symmetry $x=0$; opens upward; narrower than $y=x^2$ (since $a=2>1$); y-intercept at $(0,5)$.

23. c does **not** affect the horizontal position of the vertex. The x-coordinate of the vertex is $x=-\frac{b}{2a}$, which depends only on a and b. Changing c shifts the graph vertically (up or down) but not horizontally.

24. $y=3x^2$ is **narrower** than $y=x^2$ because $|a|=3>1$. Both open upward with vertex $(0,0)$, but $y=3x^2$ rises more steeply.

Part G — Open-Ended (sample answers)

25. e.g. $f(x)=(x+3)^2$, axis $x=-3<0$

26. e.g. $f(x)=-x^2-1$ (always $\leq -1<0$)

27. e.g. $f(x)=5x^2$ (opens up, vertex at origin)

Part H — Extended

28. **a.** Both equations give $x=\pm 2$.
b. $2x^2-8=2(x^2-4)$. Multiplying by a constant scales the graph vertically but does not change the x-intercepts (where $y=0$).

29. Let width $=w$. Then length $=2w+3$.

$$w(2w+3)=90 \implies 2w^2+3w-90=0$$

$$w=\frac{-3+\sqrt{9+720}}{4}=\frac{-3+27}{4}=6 \text{ ft}, \quad \ell=2(6)+3=15 \text{ ft}.$$

Dimensions: 6 **ft** $\times$ 15 **ft.**

30. $\pi r^2=324 \Rightarrow r^2=\dfrac{324}{\pi}\approx 103.13 \Rightarrow r\approx \mathbf{10.2}$ ft.

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